Introduction to Ansible

Ansible Fundamentals

Agenda

- What is Ansible?
- What Ansible Automates
- A short comparison of Ansible, Chef and Puppet
- Ansible History
- Installing Ansible

What is Ansible?

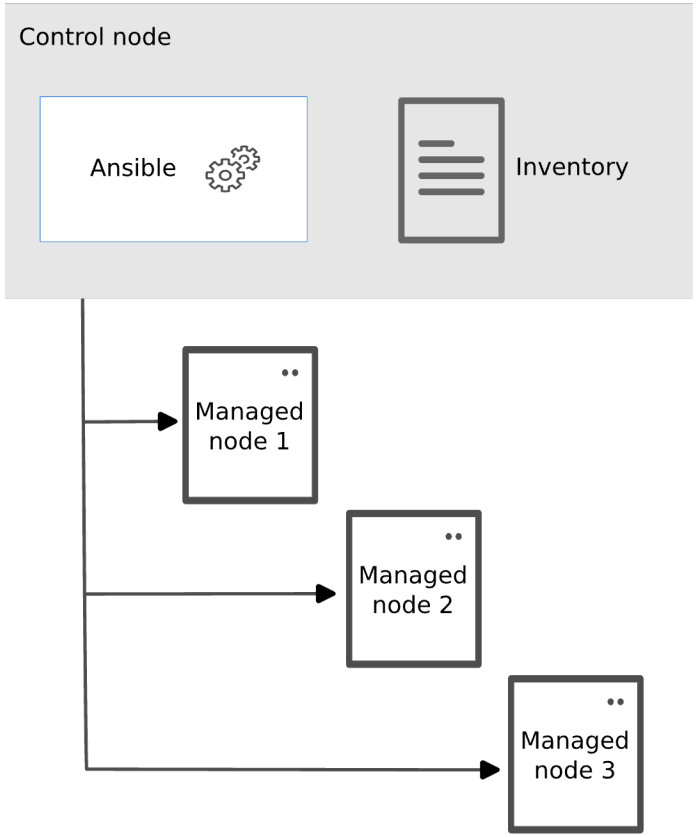
Introduction to Ansible

What is Ansible?

- **Open-Source Automation Tool**: Ansible is a free and open-source tool designed for automation tasks such as configuration management, application deployment, and infrastructure provisioning.
- **Agentless Design**: Unlike many other automation tools, Ansible operates in an agentless manner, relying on SSH (or WinRM for Windows) to communicate with target machines, eliminating the need to install additional software on those machines.
- **Playbook-Driven**: Ansible uses playbooks written in YAML (Yet Another Markup Language) to define and describe automation tasks in a human-readable format.
- **Extensibility**: With a vast library of built-in modules and the ability to create custom ones, Ansible can manage and automate diverse systems, from servers and databases to networking devices and cloud resources.

What is Ansible?

- **Control node**: A system on which Ansible is installed. You run Ansible commands such on a control node.
- **Managed node**: A remote system, or host, that Ansible controls.
- **Inventory**: A list of managed nodes that are logically organized. You create an inventory on the control node to describe host deployments to Ansible.



What Ansible Automates

Introduction to Ansible

What Ansible Automates

- Configuration Management
  - Ensure consistent configurations across servers.
  - Apply system updates and patches.
  - Manage users and groups.
- Application Deployment
  - Deploy applications to various environments.
  - Coordinate multi-tier rollouts.
  - Handle database migrations and updates.
- Infrastructure Provisioning
  - Spin up and tear down virtual machines or cloud instances.
  - Manage cloud resources, such as storage, networking, and databases.
  - Work with popular cloud providers like AWS, Azure, and Google Cloud.
- Orchestration
  - Coordinate tasks on multiple machines.
  - Handle complex deployments and workflows.
  - Manage rolling updates without downtime.

What Ansible Automates

- Network Automation
  - Configure network devices from vendors like Cisco, Juniper, and Arista.
  - Automate network setups, VLAN configurations, and firewall rules.
- Security and Compliance
  - Harden systems based on security benchmarks.
  - Ensure compliance with policies and standards.
  - Rotate passwords and manage SSH keys.
- Continuous Delivery and Integration
  - Integrate with CI/CD pipelines.
  - Automate testing and validation processes.
  - Handle artifact delivery and staging.
- Task Automation
  - Run ad-hoc commands across a group of servers.
  - Schedule routine tasks like backups and cleanups.
- Storage Management
  - Automate storage provisioning and management.
  - Handle tasks related to file systems, LVM, and RAID configurations.

What Ansible Automates

- Monitoring and Logging
  - Integrate with monitoring solutions.
  - Automate the setup of logging solutions and forwarders.
- Backup and Recovery
  - Automate backup processes for databases, applications, and systems.
  - Handle restoration processes.
- Cloud Services Integration
  - Manage services like AWS Lambda, S3, RDS, and more.
  - Automate scaling actions and resource adjustments.
- Custom Extensions
  - With its modular design, users can write custom modules to automate tasks specific to their needs.

A short comparison of Ansible, Chef and Puppet

Introduction to Ansible

Ansible vs Chef vs Puppet

	Ansible	Chef	Puppet
Design Philosophy	Agentless; relies on SSH for communication	Uses an agent-server model; requires a Chef workstation, server, and client	Agent-master model; uses a declarative approach to define the desired state
Configuration Language	Uses YAML for its playbooks	Uses a Domain Specific Language (DSL) based on Ruby called "recipes" and "cookbooks"	Puppet's own DSL
Primary Use Cases	Configuration management, application deployment, and task automation	Infrastructure automation by treating infrastructure as code	Configuration management and system automation
Learning Curve	Known for its simplicity and human-readable playbooks	Steeper due to its Ruby-based DSL, but offers a lot of flexibility and power	Requires understanding of its unique DSL; offers detailed reporting and a visual management dashboard
Community & Support	Strong open-source community; backed by Red Hat	Vibrant community with a marketplace for cookbooks; backed by Progress Software	Well-established user base; Puppet Enterprise offers commercial support

Installing Ansible

Introduction to Ansible

Control Node Requirements

- For your control node (the machine that runs Ansible), you can use nearly any UNIX-like machine with Python 3.9 or newer installed
- This includes Red Hat, Debian, Ubuntu, macOS, BSDs, and Windows under a Windows Subsystem for Linux (WSL) distribution
- Windows without WSL is not natively supported as a control node; see (more details here)

Managed Node Requirements

- The Managed node (the machine that Ansible is managing) does not require Ansible to be installed, but requires Python 2.7, or Python 3.5 - 3.11 to run Ansible-generated Python code
- The managed node also needs a user account that can connect through SSH to the node with an interactive POSIX shell.

ansible vs. ansible-core

- ansible-core
  - Minimalist language and runtime package
- ansible
  - Much larger “batteries included” package
  - Adds a community-curated selection of Ansible Collections for automating a wide variety of devices
  - Includes ansible-core
- **ansible** package is the preferred choice. **ansible-core** must be used only in limited scenarios

Instalation using Python package manager (pip)

- Install Ansible

```
$ python3 -m pip install --user ansible
```

- Upgrade Ansible

```
$ python3 -m pip install --upgrade --user ansible
```

- Confirm your installation

```
$ ansible --version
```

Instalation using OS prefered way

- You can install ansible using a direct way supported by your OS
- Installing Ansible on Fedora Linux
- Installing Ansible from EPEL
- Installing Ansible on OpenSUSE Tumbleweed/Leap
- Installing Ansible on Ubuntu
- Installing Ansible on Debian
- Installing Ansible on Arch Linux
- Installing Ansible on Windows

Check Ansible Running

Demo